

NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
CSPC 204/ITPC 204 – Artificial Intelligence and Soft Computing (CS/IT/M&C)

FM:15

1. Describe the possibility of iterative deepening search performing much worse than depth first search with a search space example. 5M
2. Given the 8-puzzle shown in following figure, compare the performance of simple hill-climbing and steepest ascent simple hill-climbing algorithm with the Manhattan distance heuristic to find a path to the goal state. 5M

1	2	3
4	8	
7	6	5

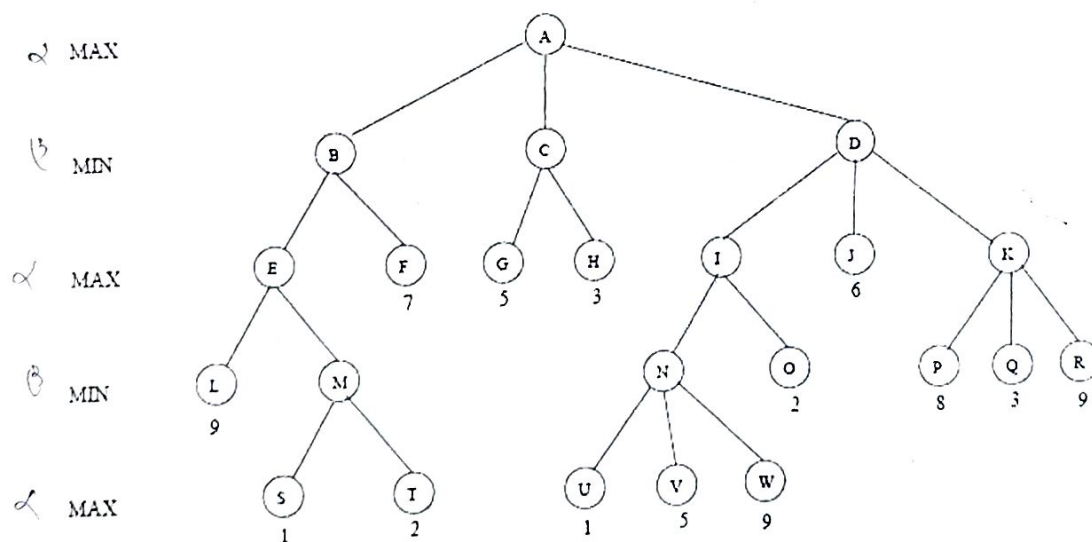
Initial state

1	2	3
4	5	6
7	8	

Goal state

3. 5M

Consider the following game tree.



Assume that the children of each node are evaluated from left to right.

- (a) Use the MINIMAX to compute the value of each node in the game tree.
- (b) List the nodes that are evaluated by ALPHA-BETA.
- (c) Redraw the game tree such that ALPHA-BETA performs the *minimum* number of evaluations (i.e., the *best case* for ALPHA-BETA). How many nodes are evaluated by ALPHA-BETA in the resulting game tree?
- (d) Redraw the game tree such that ALPHA-BETA performs the *maximum* number of evaluations (i.e., the *worst case* for ALPHA-BETA). How many nodes are evaluated by ALPHA-BETA in the resulting game tree?